S&P 500 Factor Investing

- Group participants: add names before submission.

- Assignment setting adapted from BIST100 to S&P 500.

- Data window: all available observations through May 2026.

- Required factors: ROE, P/E, momentum, and trend.

Paper Summary

- Han, Zhou, and Zhu propose a trend factor using moving averages across multiple horizons.

- Normalized moving averages combine short-term reversal, intermediate momentum, and long-term reversal information.

- The paper forms top-minus-bottom portfolios from expected returns predicted by trend signals.

- Our project keeps the idea but adapts the index input to ^GSPC.

Data And Signals

- Stocks: current S&P 500 constituent panel from Tiingo data.

- Benchmark and trend-regression index: Yahoo Finance ^GSPC.

- Fundamentals are joined by public availability date to avoid look-ahead bias.

- All factors are month-end, winsorized, z-scored, and lagged before trading.

Backtest Design

- Initial cash: 1,000,000.

- FixedCashSizer: 100,000 per trade.

- Commission: 0.

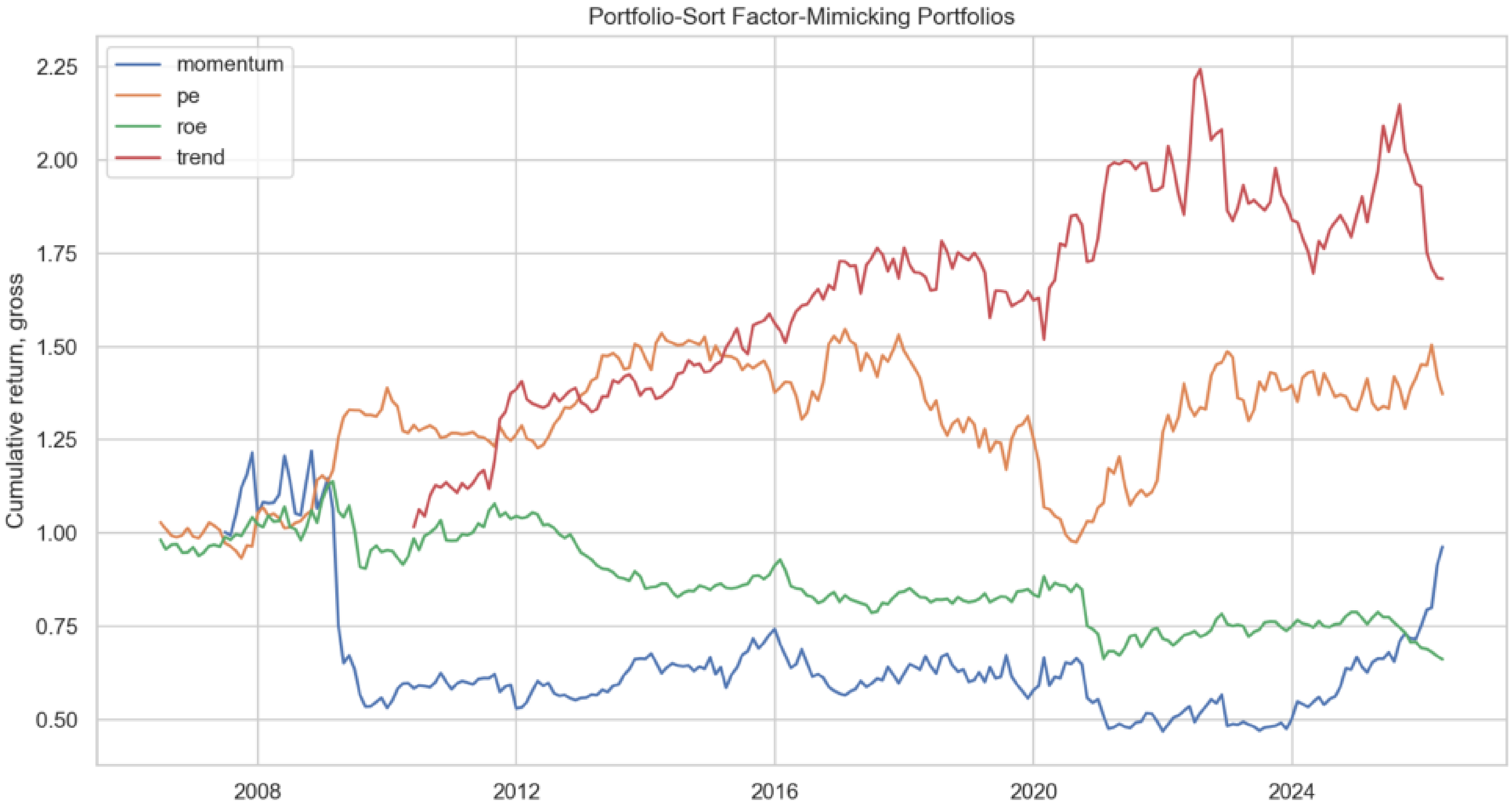
- Base strategy: long-only top 10 composite factor stocks, monthly rebalance, market orders.

- Improved 1: same base strategy with expanding no-lookahead trend regression.

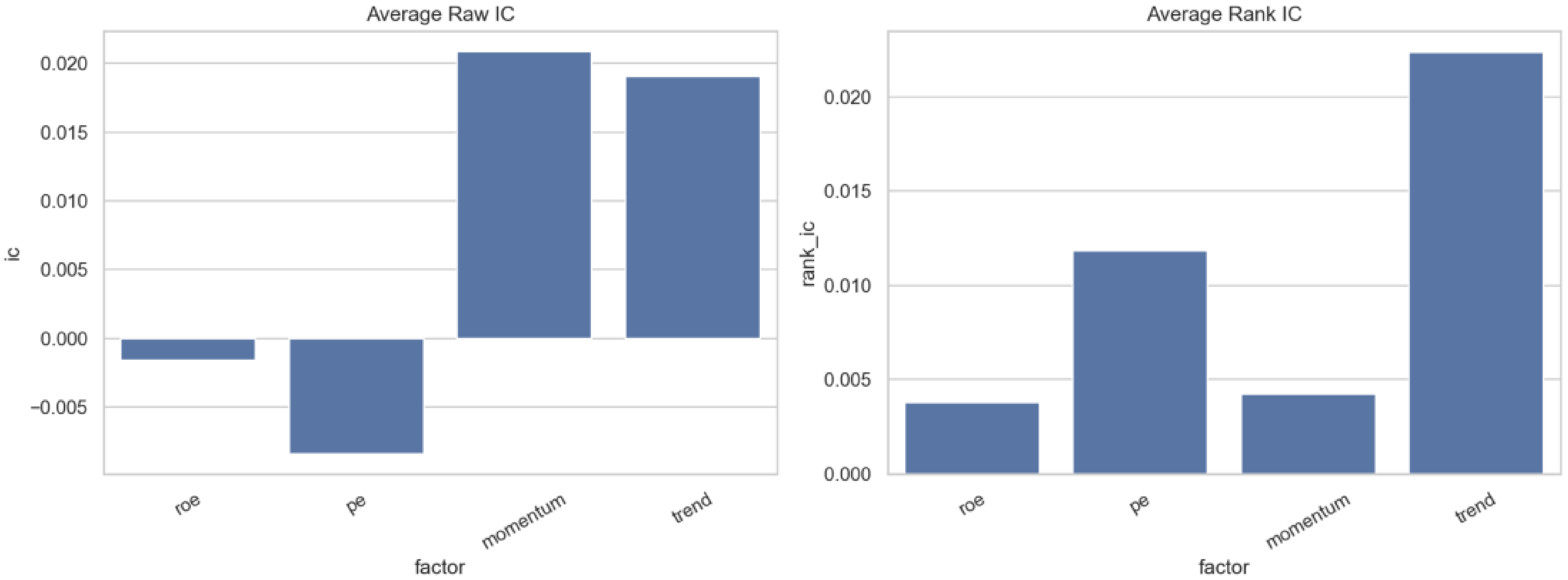
- Improved 2: improved 1 plus 10% stop-loss and 20% take-profit.

- Improved 3: improved 2 plus rolling rank-IC factor weights with shrinkage and caps.

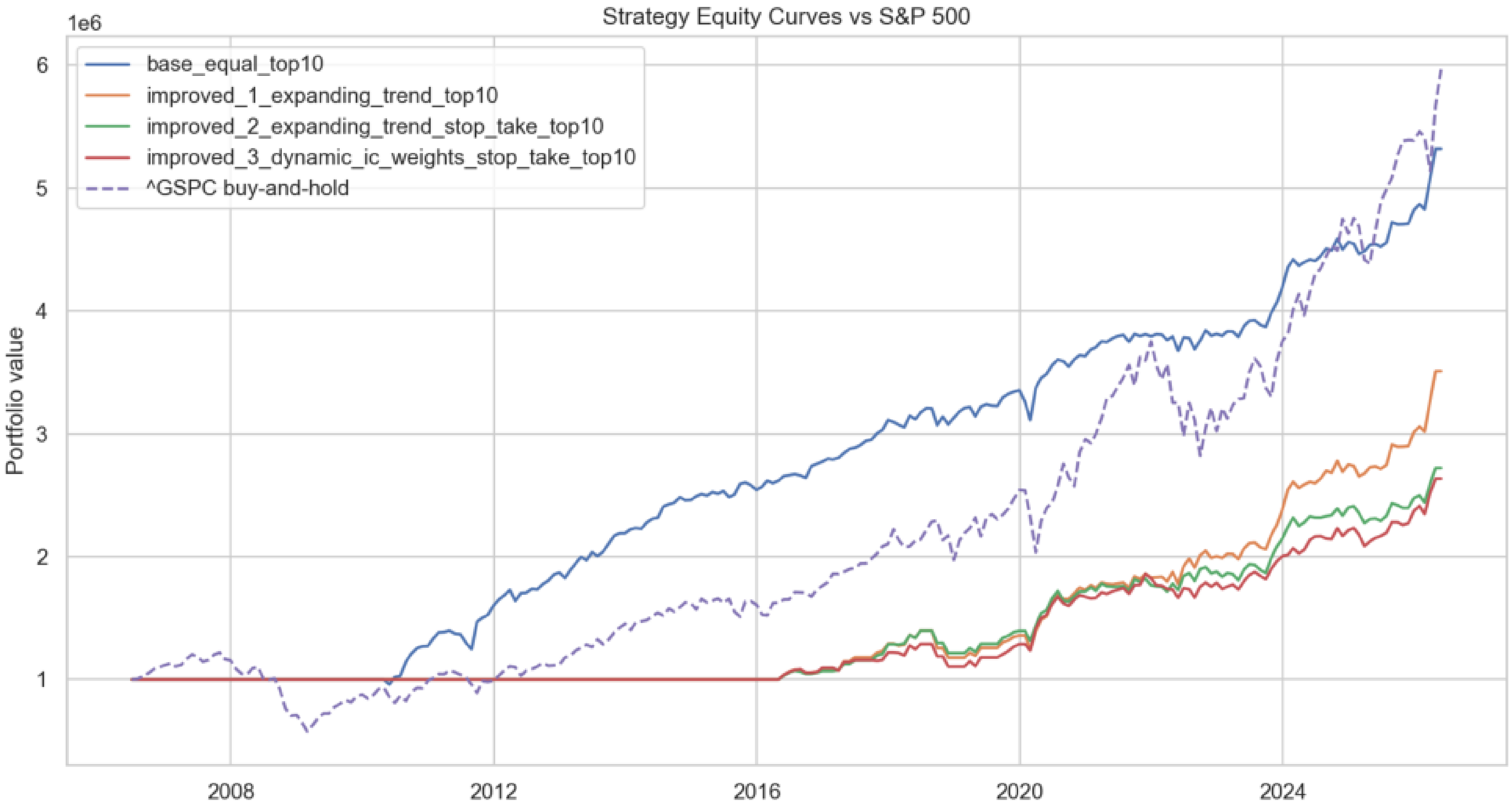
Factor Portfolio Performance



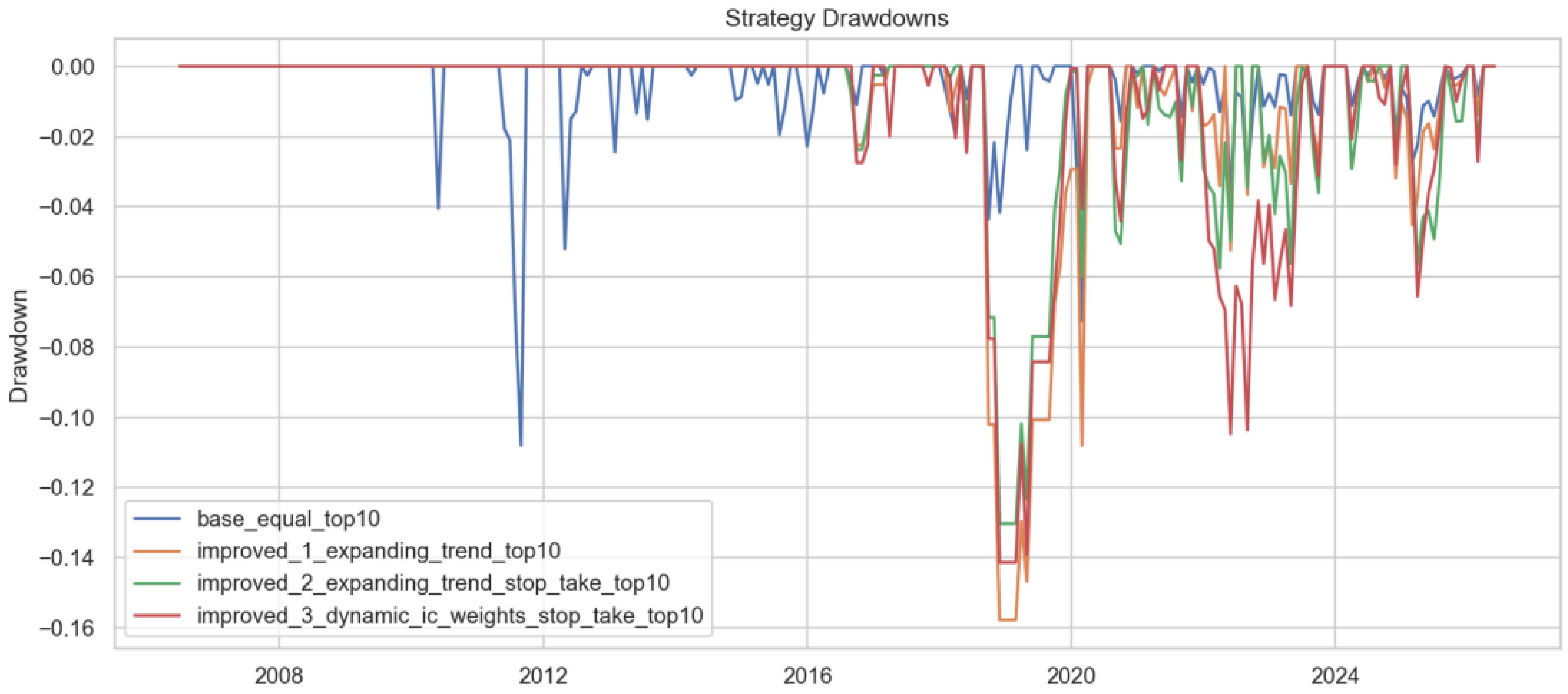
Information Coefficients



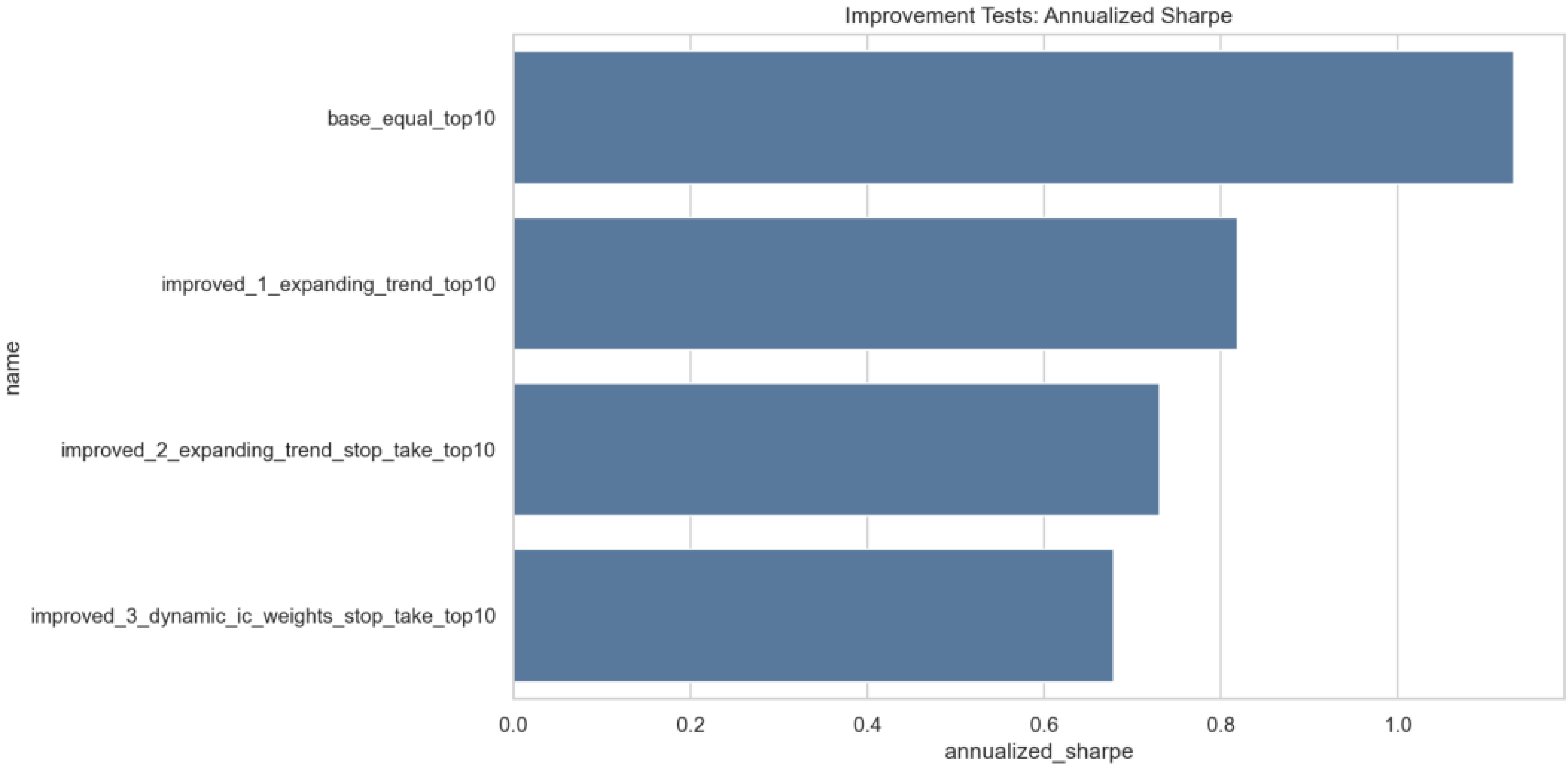
Strategy vs Benchmark



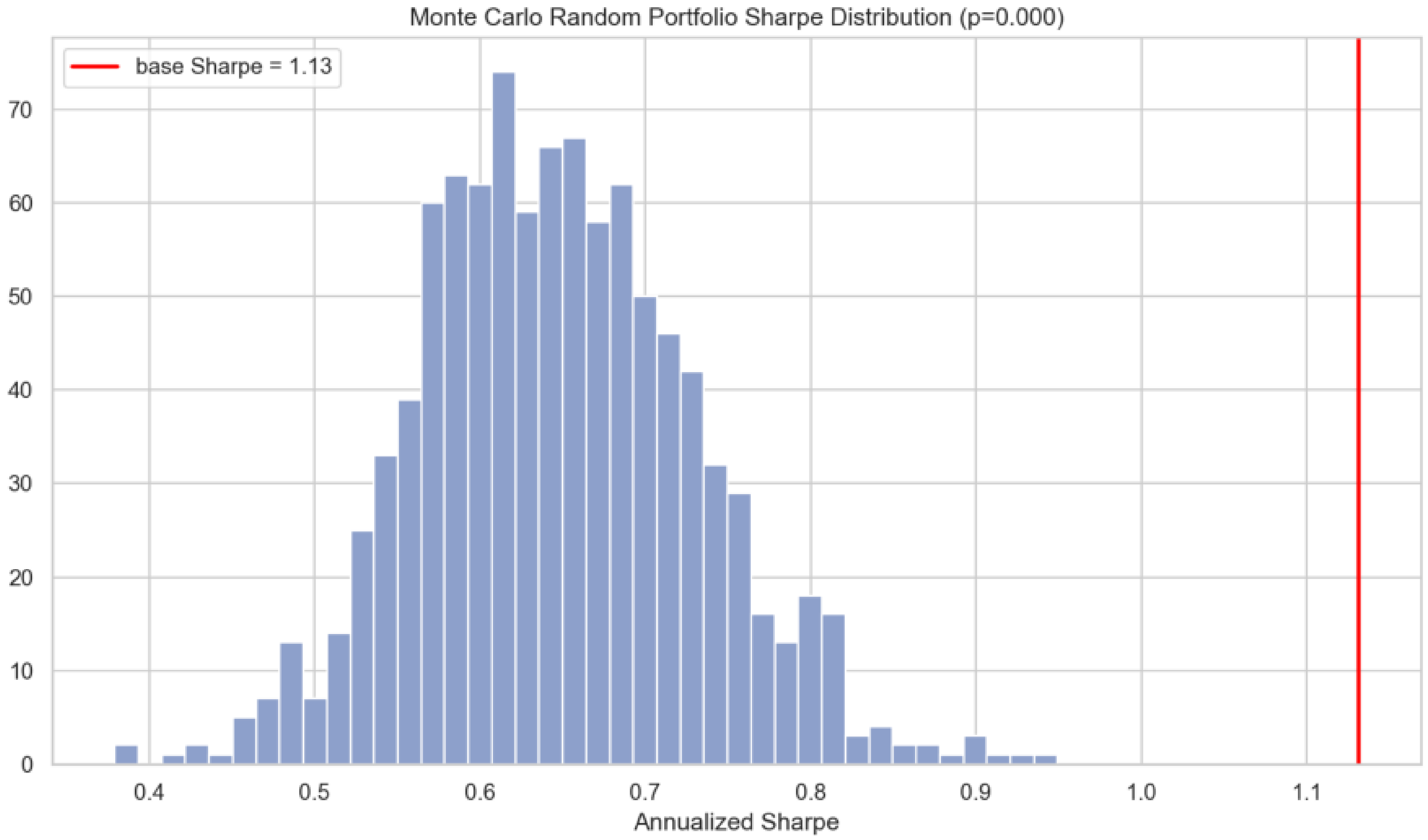
Drawdowns


Strategy Drawdowns
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
−0.12
−0.14
−0.16
Drawdown
2008
2012
2016
2020
2024
base_equal_top10
improved_1_expanding_trend_top10
improved_2_expanding_trend_stop_take_top10
improved_3_dynamic_ic_weights_stop_take_top10

Improvement Tests



Monte Carlo Robustness



Backtrader Results

- Base Backtrader final value: $5,778,291; Sharpe: 1.20.

- Improved 3 Backtrader final value: $2,643,354; Sharpe: 0.82.

- Settings: initial cash 1,000,000; FixedCashSizer 100,000; commission 0.

- Improved 2 and improved 3 use daily Backtrader stop/take threshold checks with market exits.

- Saved outputs include orders, trades, positions, equity curves, and metrics.

Main Results

- Base strategy final equity: $5,319,947; Sharpe: 1.13.

- Improved 3 strategy: improved_3_dynamic_ic_weights_stop_take_top10; final equity: $2,635,221; Sharpe: 0.68.

- Monte Carlo p-value for base Sharpe: 0.0000.

- Monte Carlo p-value for improved 3 Sharpe: 0.1710.

- Improved 3 annualized alpha vs ^GSPC: 5.43%.

- Interpret results cautiously because the universe has survivorship bias and costs are ignored.

Conclusion

- The factor process is economically interpretable and reproducible.

- The strongest evidence comes from comparing base, improvement, walk-forward, and random-portfolio tests.

- We would not present this as real-money ready without survivorship-bias-free data and realistic costs.

- Future work: historical S&P membership, transaction costs, slippage, and stronger out-of-sample validation.